



Activity File: Metasploit Exploitation

- In this activity, you will continue to play the role of a pen tester conducting an engagement on MegaCorpOne.
- Through our previous Nmap scan, we discovered several exploitable services.
- MegaCorpOne has given you permission to exploit any vulnerable service you find, as long as you document your actions.
- You are tasked with using the Metasploit framework to obtain a reverse shell on the remote host using an exploit module.

Note the following:

- This activity might be slightly challenging, and you will be tasked with attempting multiple exploits.
- Some exploits may succeed and some may fail, similar to a real penetration test.
- You'll have extra time to work through this activity.
- Use the Metasploit cheat sheet to determine the appropriate commands to run.
- Use the template outlined later in this activity file to keep notes, because you'll use this information in later activities.

 **Reminder** - Don't forget to save your findings, as you will add them to your report on Day 4!

The following image shows the results of the Nmap scan that you ran in a previous activity:

```
(root@kali)~# nmap -sV 172.22.117.150
Starting Nmap 7.92 ( https://nmap.org ) at 2022-01-13 16:23 EST
Stats: 0:00:45 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 95.65% done; ETC: 16:24 (0:00:01 remaining)
Nmap scan report for 172.22.117.150
Host is up (0.031s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:15:5D:02:04:10 (Microsoft)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 66.74 seconds

(root@kali)~#
```

- :warning: **Heads Up:** Metasploit will sometimes choose a different default network adapter, which will cause exploits to not fully establish a connection back to Ubuntu. If you encounter this, set the default LHOST via `setg LHOST 172.22.117.100`

Instructions

Your task is to use Metasploit to exploit as many services as possible within the time limit.

- To determine what exploit to attempt, use the Nmap scan results shown in the previous image to search for potential exploits.
 - **Hint:** Search for keyword values such as `vsftpd`, `smtp_enum`, and `ssh_login`.
- Once you have chosen an appropriate exploit, select (use) it.
- View and configure the required options.
- Attempt the exploit, and determine if it was successful or not.

For each exploit you attempted (failed or successful), document the following:

- **Exploit:**
- **Host IP address:**
- **Port:**
- **Service name:**
- **Service version:**

- **Exploit outcome:**

For example:

- **Exploit:** exploit/multi/http/apache_mod_cgi_bash_env_exec/
- **Host IP address:** 172.17.0.4
- **Port:** 80
- **Service name:** HTTP
- **Service version:** Apache 2.1.3
- **Exploit outcome:** Success

For attribution purposes, it's **extremely** important to document **every** exploit attempt, even if it fails. You may attempt exploitation as many times as you'd like.
